

AES-128 hardware encryption accelerator

Experiment 1 - Beginner | V-SPACE FPGA Build Challenge 2026

Team	Beyond Boolean
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Board / device	PYNQ-Z2 / xc7z020clg400-1
Tool / clock constraint	Vivado 2025.1.1 / 125 MHz (8 ns)

1. Objective

Implement iterative AES-128 encryption in programmable logic and observe a retained 128-bit ciphertext through VIO/JTAG. A key-expansion register change addresses the archived setup-timing failure while preserving the interface and cycle behavior. The original board demonstration and repaired candidate have distinct evidence.

2. Block Diagram

AES-128 hardware architecture

PYNQ-Z2 | iterative AES core with Vivado VIO control

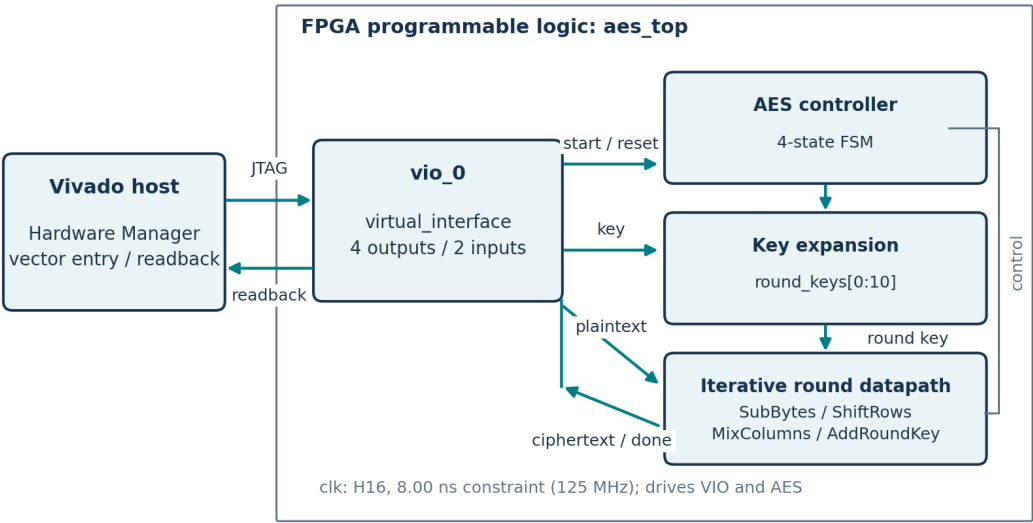


Diagram prepared from the recovered aes_top.v and aes_core.v.
Final AES round bypasses MixColumns. No AXI/PS overlay is present in this archived design.

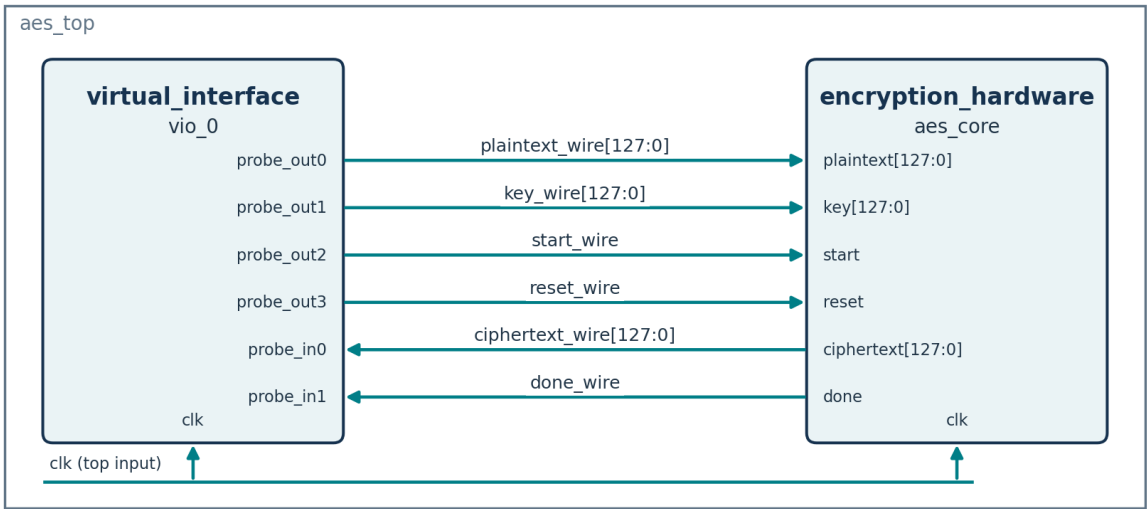
Conceptual drawing grounded in the original RTL, not a native Vivado block-design capture. The repaired core keeps this top-level architecture and adds a previous-key register internally. No AXI DMA or PS block is instantiated.

3. RTL Design

Source / instance	Role
aes_top.v	Common board clock, virtual_interface (vio_0), encryption_hardware (aes_core)
aes_core.v	Key expansion; SubBytes, ShiftRows, MixColumns and AddRoundKey; four-state control
vio_0	Four control/data outputs and two monitored inputs
pynq.xdc	H16, LVCMOS33, unchanged 8 ns constraint

Structural RTL schematic: aes_top

Exact module instances, port directions and internal wire names from aes_top.v



Source-derived structural diagram; this is not a Vivado elaborated or synthesized netlist capture.

All six data/control connections and the common clock are shown. Bus widths: plaintext/key/ciphertext = 128.

Source-derived top connectivity drawing. It represents actual port directions and widths, not a Vivado netlist GUI capture. The same top wiring applies to both revisions.

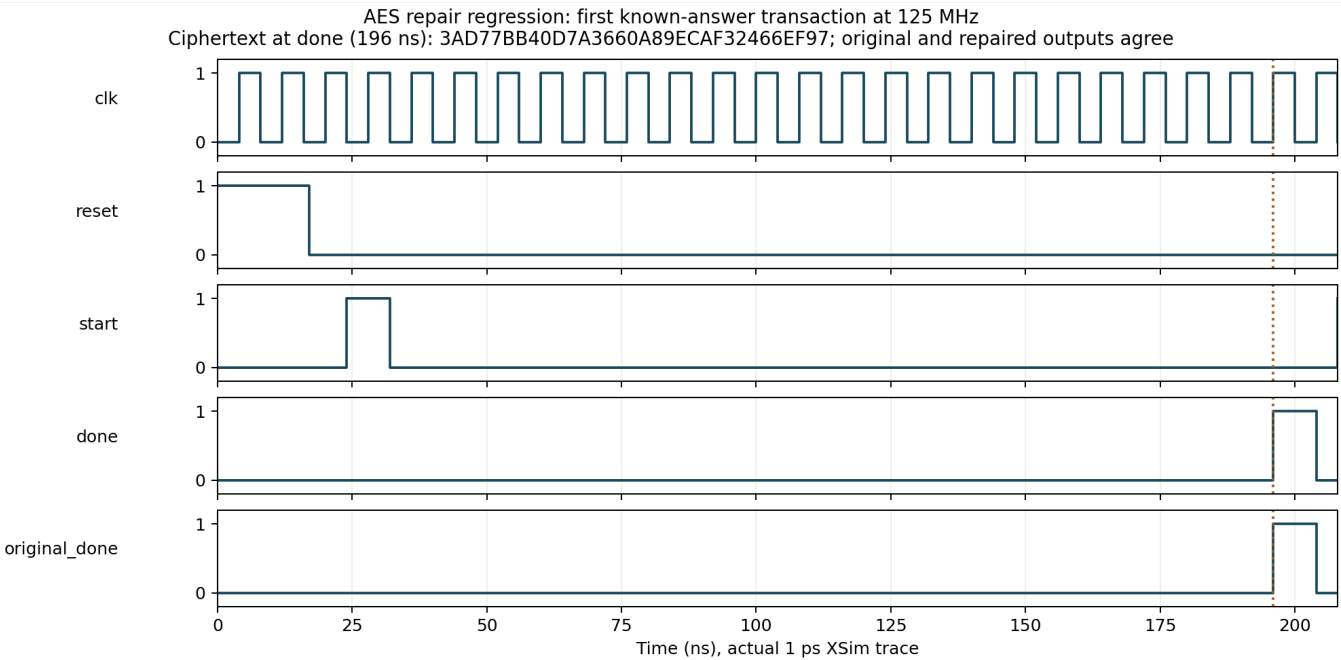
`S_IDLE` captures key/plaintext and performs the initial XOR. `S_KEY_EXP` stores ten expanded round keys. `S_ROUNDS` reuses the round datapath; round ten omits MixColumns. `S_FINISHED` registers ciphertext and asserts done, returning to IDLE. Completion is 21 clock intervals after an accepted start; done lasts one clock. Held start retriggers in IDLE.

The repair reads a dedicated 128-bit `previous_key` register, loaded on start and updated for each expansion, instead of selecting `round_keys[k_idx-1]`. It still writes the original round-key bank. Original RTL remains in RTL; candidate RTL and a source diff are in `Timing_Repair/`.

4. Simulation Results

Original: the preserved 100 MHz testbench/transcript passes one known-answer vector. Its historical waveform photo and native WDB remain in Simulation/. Repair: Vivado XSim 2025.1.1 independently reran the supplied 125 MHz equivalence regression on 10 October 2026 and reached all final PASS markers.

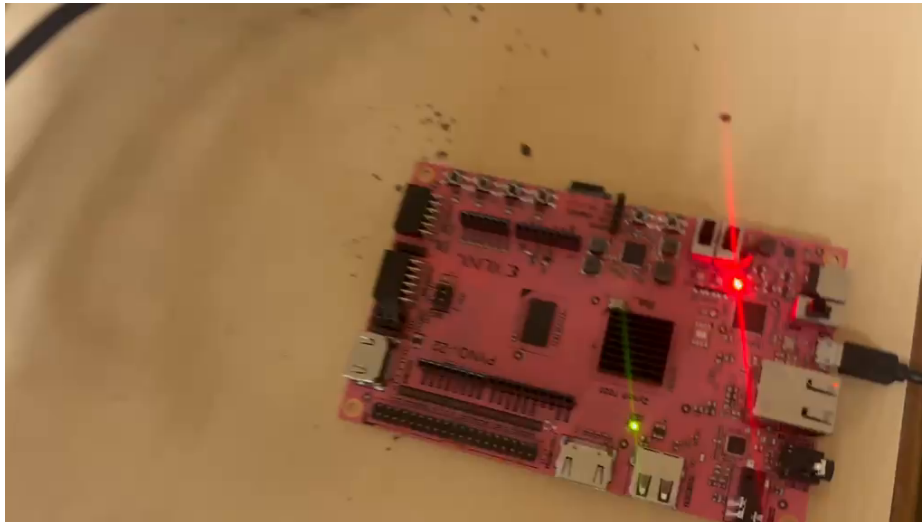
Signal	First known-answer vector
Key	2B7E151628AED2A6ABF7158809CF4F3C
Plaintext	6BC1BEE22E409F96E93D7E117393172A
Ciphertext	3AD77BB40D7A3660A89ECAAF32466EF97



Scientific plot of the actual new XSim VCD, not a fabricated simulator screenshot. Full trace: Timing_Repair/Validation/regression.vcd. Original/repaired done and ciphertext events agree throughout the regression.

104 key/plaintext/reference triples were independently recomputed with AES-ECB on individual 16-byte blocks. The paired testbench checks exact ciphertexts, cycle-by-cycle output equivalence, 21-cycle completion, one-cycle done, retention, input changes while busy, asynchronous mid-encryption reset, restart and repeated encryption with held start. Behavioral simulation does not measure routed propagation delay.

5. Hardware Implementation - original physical evidence



Unchanged powered PYNQ-Z2 frame supplied in AESRepair.zip. The original board_setup.jpg is also retained. The normal PYNQ power/boot setup and USB/JTAG are required; standalone PL evidence does not establish independent PS initialization.

Name	Value	Activity	Direction	VIO
ciphertext_wire[127:0]	[H] 3AD7_78B4_0D7A_3660_A89E_CAF3_2466_EF97		Input	hw_vio_1
done_wire	[B] 0		Input	hw_vio_1
key_wire[127:0]	[H] 287E_1516_28AE_D2A6_ABF7_1588_09CF_4F3C		Output	hw_vio_1
plaintext_wire[127:0]	[H] 6BC1_BEE2_2E4D_9F96_E93D_7E11_7393_172A		Output	hw_vio_1
reset_wire	[B] 0		Output	hw_vio_1
start_wire	[B] 0		Output	hw_vio_1

```
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 0 [get_hw_probes reset_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes reset_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 1 [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
set_property OUTPUT_VALUE 0 [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
commit_hw_vio [get_hw_probes start_wire] -of objects [get_hw_vios -of objects [get_hw_devices xc7z020_1] -filter {CELL_NAME=="virtual_interface"}]]
```

Unchanged original recording frame: programmed target and full key/plaintext/ciphertext visible. done=0 is consistent with a one-clock done pulse and retained ciphertext. This frame documents the original demonstration, not a board test of repaired programming files.

The older low-resolution hardware_output.jpg preview and programmed-VIO screenshot remain untouched. The repaired BIT/LTX pairs are separately supplied; no physical execution of either repaired pair is claimed.

5. Hardware Implementation - build results

Metric	Original archived build	Supplied repair / checkpoint recheck
Clock constraint	125 MHz / 8 ns	125 MHz / 8 ns
Setup WNS / TNS	-0.104 / -0.276 ns	1.490 / 0.000 ns
Setup failing endpoints	4	0
Hold slack / failing endpoints	0.028 ns / 0	0.033 ns / 0
Pulse-width slack	See original report	2.750 ns
Integrated LUTs / FF	3222 / 3912	2860 / 4043
BRAM / DSP	0 / 0	0 / 0
Bus skew	Original report retained	Four MET; minimum 6.642 ns
Unconstrained internal endpoints	Original report retained	0

The supplied repaired checkpoint was opened in Vivado 2025.1.1 and timing, utilization, bus skew, DRC and methodology were regenerated. Its regenerated bitstream configuration payload matches the supplied BIT exactly; the header timestamp differs. Regenerated probe JSON matches the supplied LTX. This establishes checkpoint/programming-file correspondence, not physical board operation.

A separate complete source reproduction passed XSim, synthesis, routing, explicit post-route optimization and export gates. Its measured setup/hold/pulse-width slacks are 1.490 / 0.033 / 2.750 ns. The matched output set is Timing_Repair/Reproduced_Outputs/. These measurements are separate from the supplied checkpoint table.

Warnings remain: five DRC Warning checks (three PDCN-1569 and one RTSTAT-10 inside vendor dbg_hub, plus ZPS7-1 for absent PS7); four LUTAR-1 methodology Warning checks concern asynchronous debug-hub FIFO resets. No Error/Critical Warning checks are listed. Static timing does not remove potential reset hazards. Do not describe the build as warning-free or independently bootable.

6. Applications

An iterative AES block can support an embedded cryptographic coprocessor or laboratory teaching platform. This VIO-controlled experiment verifies block encryption. Production use needs suitable interfaces, buffering, secure key handling, an encryption mode and authentication; those capabilities are future work.

7. Conclusion and Future Scope

The original design has authentic single-vector simulation and physical demonstration evidence but retains its negative 125 MHz setup slack. The separate key-expansion repair passes an extended functional equivalence regression and the supplied routed checkpoint closes static timing under the unchanged clock constraint. Original technical assets and measurements remain preserved.

Physical verification of the repaired files and review of final video content/public access remain pending. Test the repaired pair on the actual board, retain the build identity and result screenshots, and review PS initialization/debug-IP warnings. The supplied final demonstration URL is recorded in Video_Link.txt. Its presence does not establish all required presentation sections or repaired-board execution.

Reproducibility

Original native source: FPGA_Project_Files/Native_Project/Project1_AES_Vivado_Source.zip. Repaired native source: Project1_AES_Repaired_Vivado_Source.zip in the same folder; extract the complete layout and open AESRepair/build_repaired/aes125_repaired.xpr. Clean repair reproduction uses Timing_Repair/BUILD_REPAIRED.tcl in a new short writable folder. Each BIT/LTX pair is tied to its own checkpoint and manifest.

Provenance and preserved records

Hardware frames come unchanged from the supplied ZIP. Architecture/connectivity drawings are source-based illustrations. Original RTL, constraints, IP, testbench, reports, native project, programming files, WDB, transcript and photographs are preserved. The previous report PDF is retained in Documentation/Archive. New validation logs/reports are labeled by origin; no fresh board execution is claimed.

AI assistance was used to prepare documentation and validation tooling. This disclosure does not assign a percentage of HDL authorship. Organizer file requirements are mapped in SUBMISSION_READINESS.md. The original organizer PDF lists 23 September 2026 as its deadline; current deadline/naming must be confirmed separately.